

STM32 Development Board with Micro USB

PCB Design Project | Tool: KiCad | 2-Layer PCB

Design Tool	KiCad
Board Type	2-Layer PCB
Power Supply	USB 5V (VBUS) → LM1117 → VCC

1. Project Overview

Arduino-compatible MCU development board with Micro USB interface, LM1117 5V regulation, dual 15-pin headers, and onboard LED+reset. Designed in KiCad with ATmega328P-M for embedded development use.

2. Key Components

- ATmega328P-M – MCU (Arduino-compatible)
- USB_B_Micro – Micro USB connector
- LM1117DT-5.0 – 5V LDO Regulator
- Resonator (3-pin SMD)
- 2x 15-pin Headers (Arduino pin-compatible)
- Decoupling Capacitors, Pull-up Resistors
- LED (Power indicator)
- Reset Push Switch

3. Design Features

- Micro USB for power and programming
- Arduino-compatible 15-pin dual header layout
- LM1117 onboard 5V regulation from USB VBUS
- SMD resonator for clock stability
- Compact 2-layer PCB design
- Reset button for manual MCU reset

4. Signal / Functional Flow

- [1] Micro USB Connector (VBUS / D+ / D-) ↓
- [2] LM1117DT-5.0 – Voltage Regulator ↓
- [3] ATmega328P-M – MCU ↓
- [4] SMD Resonator – Clock Source ↓
- [5] 15-pin Dual Headers – GPIO / Power Expansion ↓
- [6] LED – Power Indicator ↓
- [7] Reset Switch

5. Resume / Portfolio Description

- Designed Arduino-compatible development board in KiCad featuring ATmega328P-M, Micro USB interface, LM1117DT-5V regulator, dual 15-pin expansion headers.
- Board includes SMD resonator, decoupling network, power LED, and reset switch; Gerber files generated for fabrication.

6. Short Description (Portfolio / GitHub – ≤350 chars)

Arduino-compatible MCU development board with Micro USB interface, LM1117 5V regulation, dual 15-pin headers, and onboard LED+reset. Designed in KiCad with ATmega328P-M for embedded development use.

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